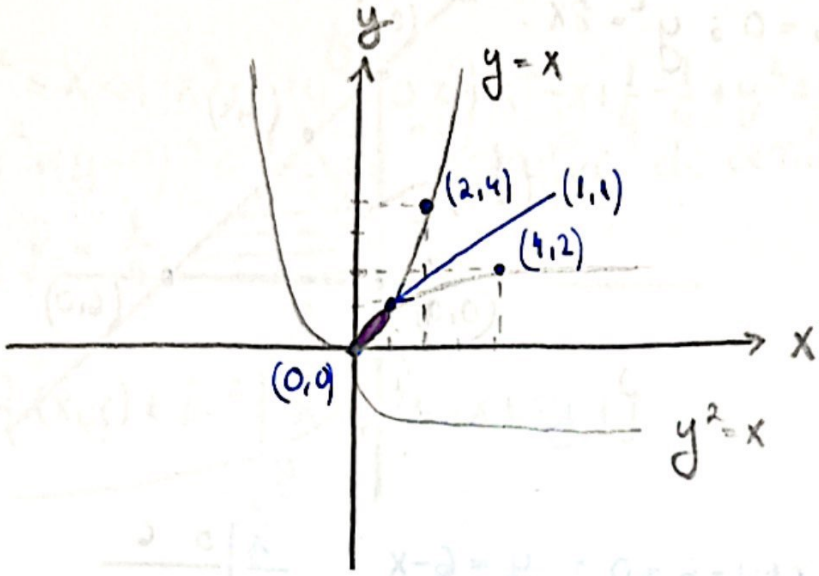
Integrale duble

• Să se reprezinte grafic următoarele domenii (determinând și coordonatele punctelor de intersecție):

(1.) D este domeniul mărginit de parabolele $y=x^2$ și $y^2=x$



• pt $y=x^2$: $x_{vf}=0$; $y_{vf}=0$; $x=1 \Rightarrow y=1$; $x=2 \Rightarrow y=4$ ....

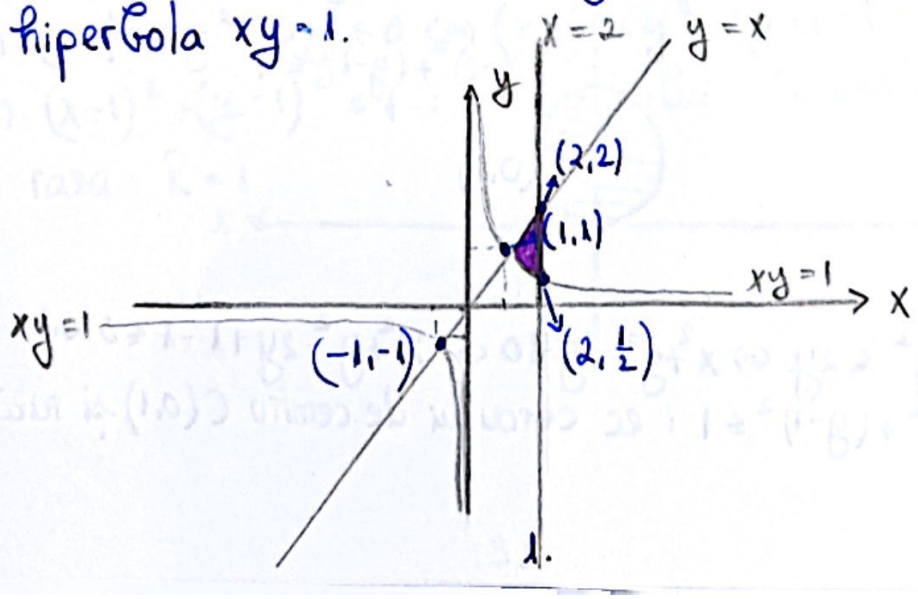
• pt $y^2=x$: $x_{vf}=0$; $y_{vf}=0$; $x=1 \Rightarrow y=1$; $y=2 \Rightarrow x=4$ ...

pct de intersecție : $y=x^2$; $y^2=x \Leftrightarrow y=\sqrt{x} \Rightarrow x^2=\sqrt{x}\uparrow^2 \Rightarrow$

$\Rightarrow x^4=x \Leftrightarrow x^4-x=0 \Leftrightarrow x(x^3-1)=0 \Leftrightarrow x=0$ sau $x=1$

pt $x=0 \Rightarrow y=0$; pt $x=1 \Rightarrow y=1 \Rightarrow (0,0)$; $(1,1)$ pct de int.

(2.) D este domeniul mărginit de dreptele $x=2$; $y=x$ și hiperbola $xy=1$.

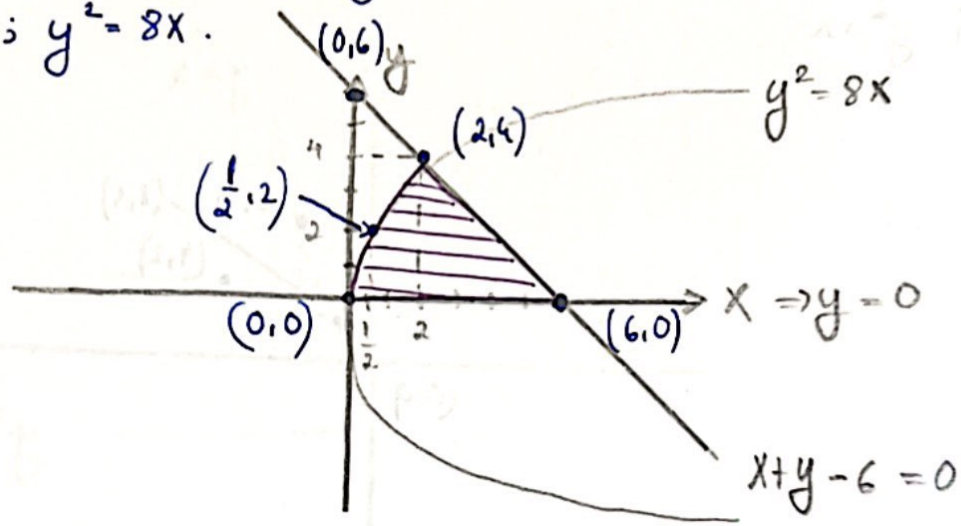


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pct de intersectie:

- $x=2$; $y=x \Rightarrow (2,2)$
- $x=2$; $xy=1 \Rightarrow \left(2,\frac{1}{2}\right)$
- $x=y$; $xy=1 \Rightarrow x^2=1 \Rightarrow x=1 \Rightarrow y=1 \Rightarrow (1,1)$

(3.) D este domeniul marginit de curbele $y=0$; $x+y-6=0$; $y^2=8x$.



- pt $x+y-6=0$: $y=6-x$

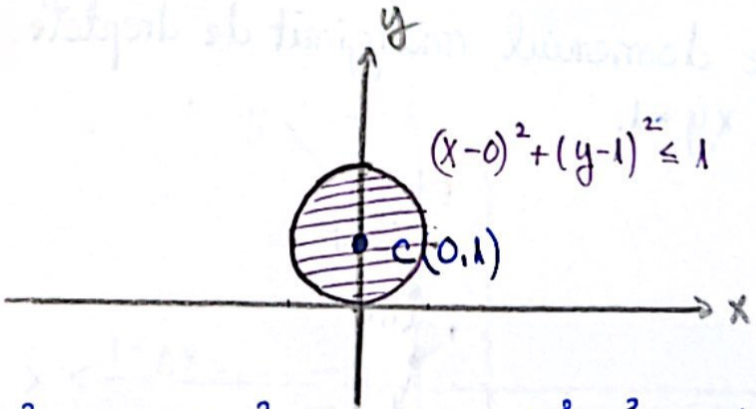
x	0	6
y	6	0

- pt $y^2=8x \Rightarrow x=\frac{1}{8}y^2$: $x_{vf}=0$; $y_{vf}=0$

$y=2 \Rightarrow x=\frac{1}{2}$; $y=4 \Rightarrow x=2$

pct de intersectie: $(0,0)$; $(6,0)$; $(2,4)$

(4.) $D=\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2\leq 2y\}$

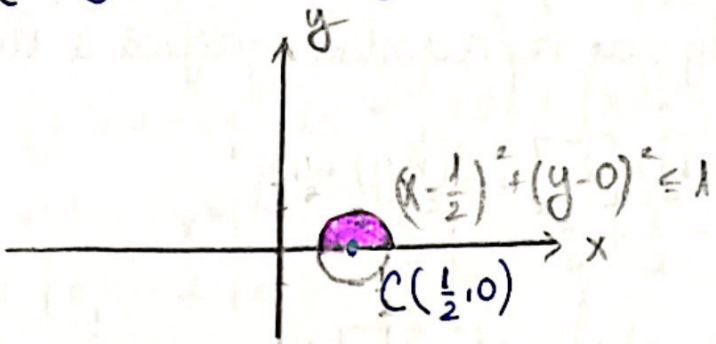


$x^2+y^2\leq 2y \Leftrightarrow x^2+y^2-2y\leq 0 \Leftrightarrow x^2+y^2-2y+1-1\leq 0 \Leftrightarrow$

$\Leftrightarrow (x-0)^2+(y-1)^2\leq 1$: ec cercului de centru $C(0,1)$ și rază $R=1$.

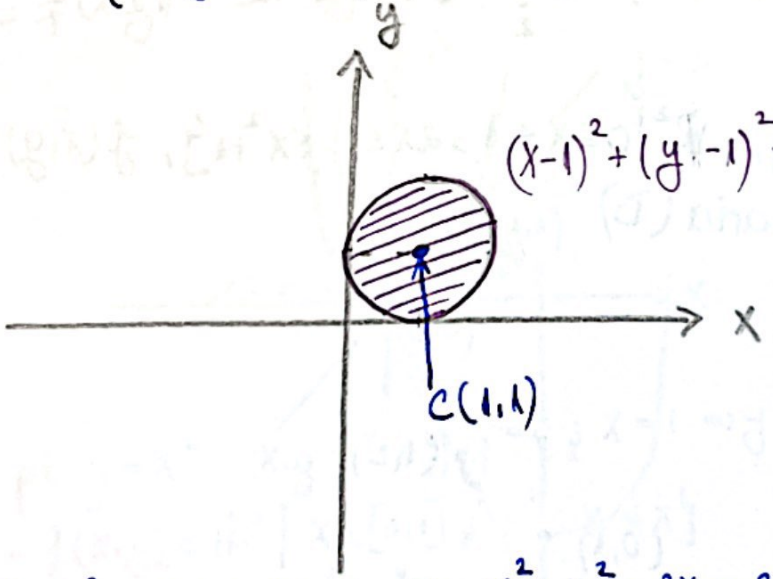
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5. $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq x, y \geq 0\}$



$x^2+y^2 \leq x \Leftrightarrow x^2-x+y^2 \leq 0 \Leftrightarrow x^2-x+\frac{1}{4}-\frac{1}{4}+y^2 \leq 0 \Leftrightarrow$
$\Leftrightarrow \left(x-\frac{1}{2}\right)^2+(y-0)^2 \leq \frac{1}{4}$: ec cercului de centru $C\left(\frac{1}{2},0\right)$
și raza $R = \frac{1}{4}$

6. $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq 2x+2y-1\}$

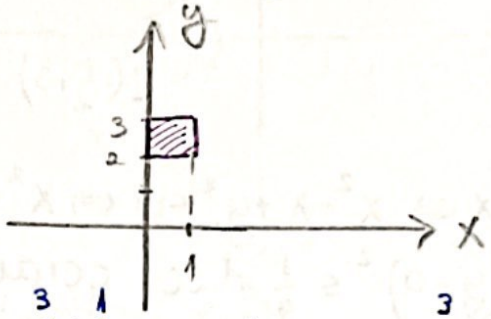


$x^2+y^2 \leq 2x+2y-1 \Leftrightarrow x^2+y^2-2x-2y+1 \leq 0 \Leftrightarrow$
$\Leftrightarrow (x-1)^2+y^2-2y \leq 0 \Leftrightarrow (x-1)^2+y^2-2y+1-1 \leq 0 \Leftrightarrow$
$\Leftrightarrow (x-1)^2+(y-1)^2 \leq 1$: ec cercului de centru $C(1,1)$
și raza $R = 1$.

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• Să se calculeze următoarele integrale duble $\iint_D f(x,y)\,dx\,dy$, cu reprezentare grafică a lui D:

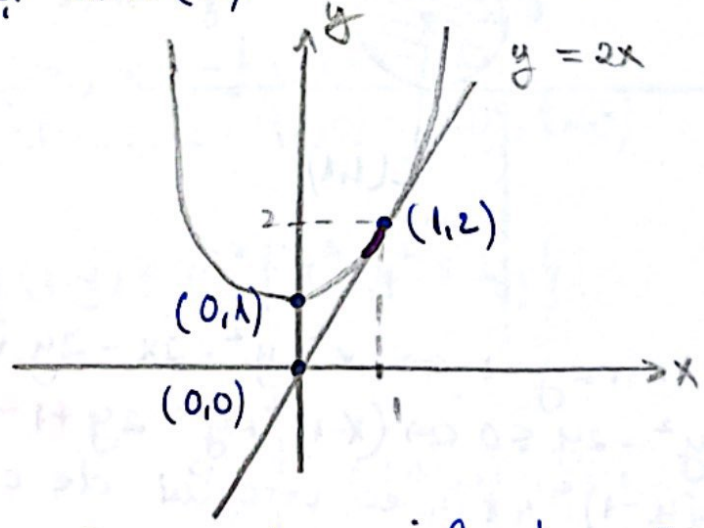
(7) $D = [0,1] \times [2,3]$, $f(x,y) = xy^2$



$$\iint_D f(x,y)\,dx\,dy = \int_2^3 \left(\int_0^1 xy^2\,dx \right) dy = \int_2^3 \left(\frac{y^2}{2} \cdot x^2 \Big|_0^1 \right) dy =$$

$$= \int_2^3 \frac{y^2}{2}\,dy = \frac{1}{2} \int_2^3 y^2\,dy = \frac{1}{2} \cdot \frac{1}{3} y^3 \Big|_2^3 = \frac{1}{6}(27-8) = \frac{19}{6}.$$

(8) $D = \{(x,y) \in \mathbb{R}^2 \mid 0 \le x \le 1 ; 2x \le y \le x^2+1\}$, $f(x,y) = x$; Calculați și aria (D)



- $2x \le y \le x^2+1 \Rightarrow$ domeniul este mărginit de dreapta $y = 2x$ și parabola $y = x^2+1$.
- pt $y = 2x$:

x	0	1
y	0	2

- pt $y = x^2+1$: $x_{vf} = 0$; $y_{vf} = \frac{-\Delta}{4a} = \frac{4}{4} = 1$

$x = 1 \Rightarrow y = 2$

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$$\iint_D f(x,y)\,dx\,dy = \int_0^1 \left(\int_{2x}^{x^2+1} x\,dy \right) dx - \int_0^1 \left(xy \Big|_{2x}^{x^2+1} \right) dx =$$

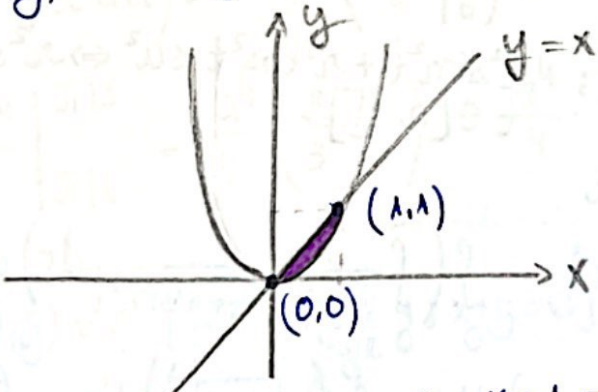
$$= \int_0^1 (x^3 + x - 2x^2)\,dx = \int_0^1 x^3\,dx + \int_0^1 x\,dx - 2\int_0^1 x^2\,dx =$$

$$= \frac{x^4}{4}\Big|_0^1 + \frac{x^2}{2}\Big|_0^1 - 2\cdot\frac{x^3}{3}\Big|_0^1 = \frac{1}{4} + \frac{1}{2} - \frac{2}{3} = \frac{1}{12}$$

$$\text{aria}(D) = \int_0^1 (x^2 + 1 - 2x)\,dx = \int_0^1 x^2\,dx + \int_0^1 1\,dx - 2\int_0^1 x\,dx =$$

$$= \frac{x^3}{3}\Big|_0^1 + x\Big|_0^1 - \cancel{2}\cdot\frac{x^2}{\cancel{2}}\Big|_0^1 = \frac{1}{3} + 1 - 1 = \frac{1}{3}$$

(9). D este domeniul mărginit de curbele $y = x$ și $y = x^2$; $f(x,y) = 3x - y + 2$; Calculați și aria(D).



- pt $y = x^2$: $x_{vf} = 0$; $y_{vf} = 0$; $x = 1 \Rightarrow y = 1$

$$D = \{(x,y) \in \mathbb{R}^2 \mid x \in [0,1];\ x^2 \le y \le x\}$$

$$\iint_D f(x,y)\,dx\,dy = \int_0^1 \left(\int_{x^2}^{x} (3x - y + 2)\,dy \right) dx = \int_0^1 \left(3xy + y^2\cdot\frac{1}{2} + 2y \Big|_{x^2}^{x} \right) dx$$

$$= \int_0^1 \left(3x^2 - 3x^3 + \frac{x^2}{2} - \frac{x^4}{2} + 2x - 2x^2 \right) dx =$$

$$= \frac{3}{2}\int_0^1 x^2\,dx + 2\int_0^1 x\,dx - 3\int_0^1 x^3\,dx - \frac{1}{2}\int_0^1 x^4\,dx =$$

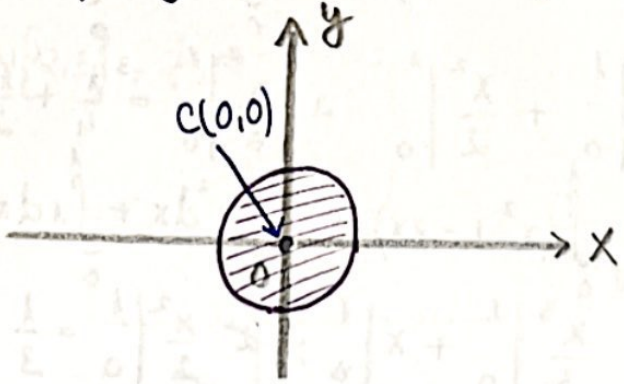
$$= \frac{\cancel{3}}{\cancel{2}}\cdot\frac{x^3}{\cancel{3}}\Big|_0^1 + \cancel{2}\cdot\frac{x^2}{\cancel{2}}\Big|_0^1 - 3\cdot\frac{x^4}{4}\Big|_0^1 - \frac{1}{2}\cdot\frac{x^5}{5}\Big|_0^1 =$$

$$= \frac{10}{2}\cdot\frac{1}{\cancel{2}} + 1 - \frac{5}{4}\cdot\frac{3}{\cancel{4}} - \frac{2}{10}\cdot\frac{1}{\cancel{10}} = \frac{10 + 20 - 15 - 2}{20} = \frac{13}{20}$$

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$$\text{aria}(D) = \int_0^1 (x - x^2)\,dx = \left(\frac{x^2}{2} - \frac{x^3}{3}\right)\Big|_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

10. $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le a^2,\ a>0,\ x \ge 0;\ y \ge 0\}$; $f(x,y) = \dfrac{1}{\sqrt{1+x^2+y^2}}$.



$x^2+y^2 \le a^2 \Leftrightarrow (x-0)^2+(y-0)^2 \le a^2$: ec cercului de centru $C(0,0)$ și rază $R = a^2$.

Folosim schimbarea de variabilă și trecem la coord. polare:

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases}; \quad r^2\sin^2 t + r^2\cos^2 t \le a^2 \Leftrightarrow r^2 \le a^2 \Leftrightarrow r \in [0,a]$$
$$t \in [0, \tfrac{\pi}{2}]$$

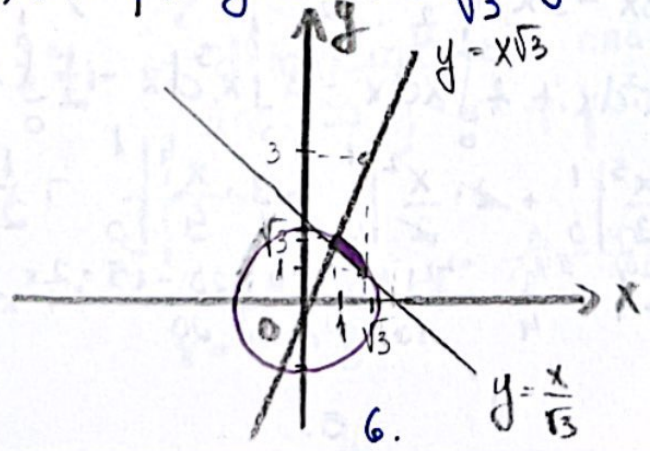
$$J(r,t) = r$$

$$\Rightarrow \iint_D f(x,y)\,dx\,dy = \int_0^{\pi/2}\left(\int_0^a \frac{r}{\sqrt{1+r^2\cos^2 t + r^2\sin^2 t}}\,dr\right)dt =$$

$$= \int_0^{\pi/2}\left(\int_0^a \frac{r}{\sqrt{1+r^2}}\,dr\right)dt = \int_0^{\pi/2}\left(\sqrt{1+r^2}\Big|_0^a\right)dt = \int_0^{\pi/2}\left(\sqrt{1+a^2}-1\right)dt =$$

$$= t\left(\sqrt{1+a^2}-1\right)\Big|_0^{\pi/2} = \frac{\pi}{2}\left(\sqrt{1+a^2}-1\right).$$

11. $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le a^2,\ 0 \le \dfrac{x}{\sqrt{3}} \le y \le x\sqrt{3}\}$; $f(x,y) = x^2+y^2$.



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$x^2+y^2 \le a^2 \Leftrightarrow (x-0)^2+(y-0)^2 \le a^2$: ec cercului de centru $C(0,0)$ și raza $R = a^2$.

Domeniul mai este mărginit de dreptele $y = \frac{x}{\sqrt{3}}$ și $y = x\sqrt{3}$.

pt $y = \frac{x}{\sqrt{3}}$:

x	1	$\sqrt{3}$
y	$\sqrt{3}/3$	1

pt $y = x\sqrt{3}$:

x	1	$\sqrt{3}$
y	$\sqrt{3}$	3

Facem schimbarea de variabilă și trecem la coord polare :

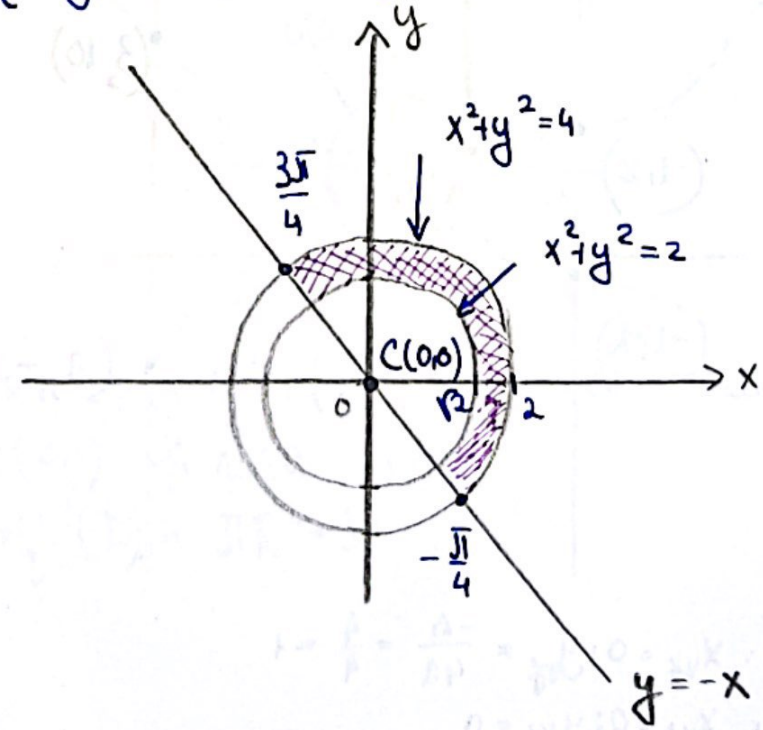
$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases}; \quad r \in [0,a]; \quad t \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]; \quad J(r,t) = r.$$

$$\Rightarrow \iint_D f(x,y)\,dx\,dy = \int_{\pi/6}^{\pi/3} \left(\int_0^a (r^2\cos^2 t + r^2\sin^2 t)\,dr \right) dt =$$

$$= \int_{\pi/6}^{\pi/3} \left(\int_0^a r^3\,dr \right) dt = \int_{\pi/6}^{\pi/3} \left(\frac{r^4}{4}\Big|_0^a \right) dt = \int_{\pi/6}^{\pi/3} \frac{a^4}{4}\,dt =$$

$$= t\frac{a^4}{4}\Big|_{\pi/6}^{\pi/3} = \frac{a^4}{4}\left(\frac{\pi}{3} - \frac{\pi}{6}\right) = \frac{a^4}{4}\cdot\frac{\pi}{6} = \frac{a^4\pi}{24}.$$

(12.) $D = \{(x,y) \in \mathbb{R}^2 \mid 2 \le x^2+y^2 \le 4,\ x+y \ge 0\}$; $f(x,y) = \sqrt{x^2+y^2}$.



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pt $2 \le x^2+y^2 \Leftrightarrow (x-0)^2+(y-0)^2 \ge 2 \Rightarrow$ ec cercului de centru $C(0,0)$ și raza $R = \sqrt{2}$.

pt $x^2+y^2 \le 4 \Leftrightarrow (x-0)^2+(y-0)^2 \le 4 \Rightarrow$ ec cercului de centru $C(0,0)$ și raza $R = 2$.

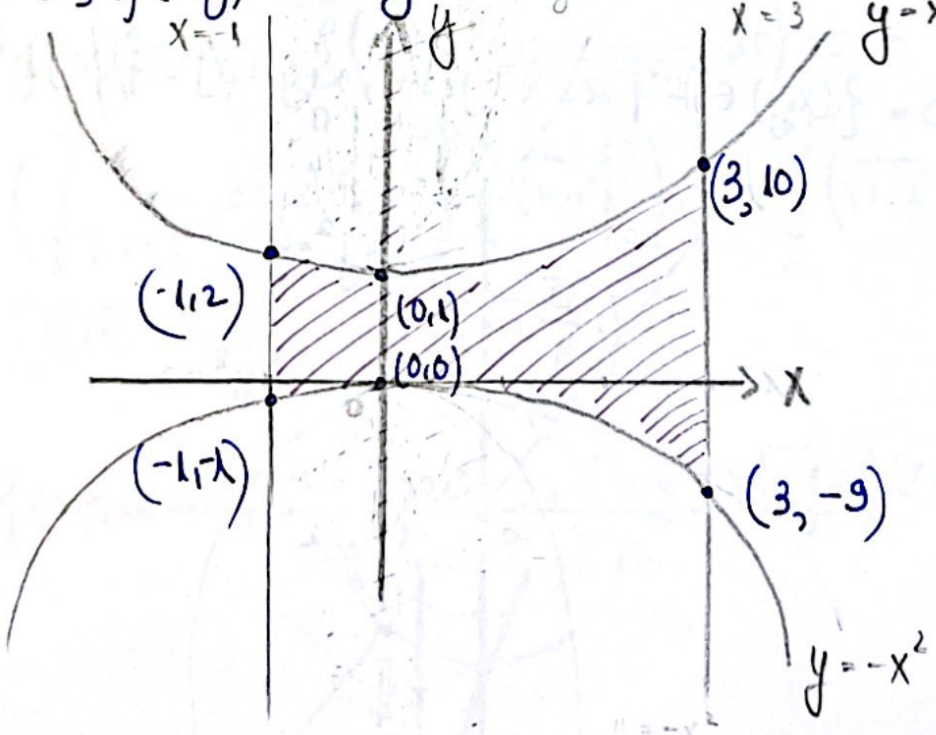
Facem schimbarea de variabilă și trecem la coordonate polare

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow r \in [\sqrt{2}, 2];\ t \in \left[-\frac{\pi}{4}, \frac{3\pi}{4}\right];\ J(r,t) = r.$$

$$\Rightarrow \iint_D f(x,y)\,dx\,dy = \int_{-\pi/4}^{3\pi/4} \left(\int_{\sqrt{2}}^{2} r^2\,dr \right) dt = \int_{-\pi/4}^{3\pi/4} \left(\frac{r^3}{3}\Big|_{\sqrt{2}}^{2} \right) dt =$$

$$= \int_{-\pi/4}^{3\pi/4} \frac{8-2\sqrt{2}}{3}\,dt = t\left(\frac{8-2\sqrt{2}}{3}\right)\Big|_{-\pi/4}^{3\pi/4} = \frac{(8-2\sqrt{2})\pi}{3}.$$

13. D este domeniul mărginit de curbele $y = x^2+1$, $y = -x^2$, $x = -1$, $x = 3$; $f(x,y) = x+3y$.



pt $y = x^2+1$: $x_{vf} = 0$; $y_{vf} = \frac{-\Delta}{4a} = \frac{4}{4} = 1$

pt $y = -x^2$: $x_{vf} = 0$; $y_{vf} = 0$

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Scriem domeniul ca $D = \{(x,y) \in \mathbb{R}^2 \mid x \in [-1,3];\ -x^2 \le y \le x^2+1\}$.

$$\iint_D f(x,y)\,dx\,dy = \int_{-1}^{3}\left(\int_{-x^2}^{x^2+1}(x+3y)\,dy\right)dx = \int_{-1}^{3}\left[\left(xy+\frac{3y^2}{2}\right)\Big|_{-x^2}^{x^2+1}\right]dx =$$

$$= \int_{-1}^{3}\left[x^3+x+\frac{3}{2}(x^2+1)^2-\left(-x^3-\frac{3x^4}{2}\right)\right]dx =$$

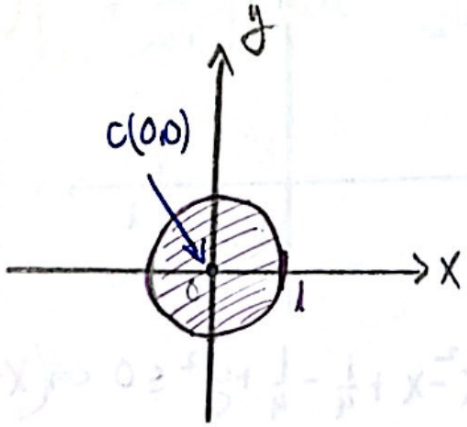
$$= \int_{-1}^{3}\left(2x^3+x+\frac{3}{2}x^4+3x^2+\frac{3}{2}+\frac{3}{2}x^4\right)dx =$$

$$= \int_{-1}^{3}\left(3x^4+2x^3+3x^2+x+\frac{3}{2}\right)dx =$$

$$= \left(\frac{3}{5}x^5+\frac{1}{2}x^4+x^3+\frac{x^2}{2}+\frac{3}{2}x\right)\Big|_{-1}^{3} =$$

$$= \frac{3}{5}\cdot 243+\frac{1}{2}\cdot 81+27+\frac{9}{2}+\frac{6}{2}-\left(-\frac{3}{5}+\frac{1}{2}-1+\frac{1}{2}-\frac{3}{2}\right) = \dots$$

(14.) $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 1\}$; $f(x,y) = e^{x^2+y^2}$. Calculați și aria (D).



$x^2+y^2 \le 1 \Leftrightarrow (x-0)^2+(y-0)^2 \le 1$: ec unui cerc de centru $C(0,0)$ și raza $R=1$

$\Rightarrow$ aria $(D) = \pi R^2 = \pi$.

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Facem schimbarea de variabilă și trecem la coord polare:

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases}; \quad r \in [0,1]; \quad t \in [0, 2\pi); \quad J(r,t) = r$$

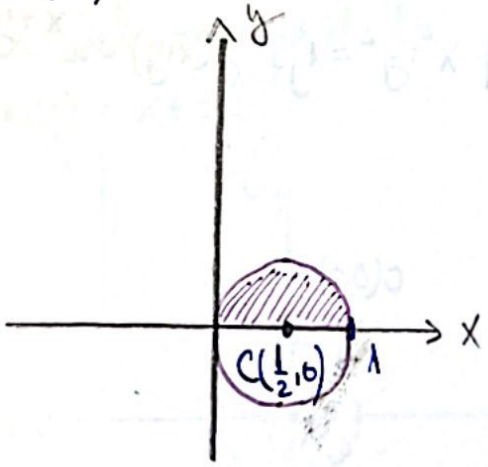
$$\iint_D f(x,y)\,dx\,dy = \int_0^{2\pi}\left(\int_0^1 r\cdot e^{r^2}\,dr\right)dt = \int_0^{2\pi}\left(\frac{1}{2}\int_0^1 2r\cdot e^{r^2}\,dr\right)dt =$$

$$= \int_0^{2\pi}\left(\frac{1}{2}\int_0^1 (e^{r^2})'\,dr\right)dt = \frac{1}{2}\int_0^{2\pi}\left(e^{r^2}\Big|_0^1\right)dt = \frac{1}{2}\int_0^{2\pi}(e-1)\,dt =$$

$$= \frac{1}{2}(e-1)\,t\Big|_0^{2\pi} = \pi(e-1)$$

15. $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le x;\ y \ge 0\}$; $f(x,y) = xy$.

Calculați și aria (D).



$x^2+y^2 \le x \Leftrightarrow x^2-x+\frac{1}{4}-\frac{1}{4}+y^2 \le 0 \Leftrightarrow \left(x-\frac{1}{2}\right)^2+(y-0)^2 \le \frac{1}{4}$:

ec cercului de centru $\left(\frac{1}{2},0\right)$ și raza $R = \frac{1}{2}$

D este un semicer de raza $\frac{1}{2}$ $\Rightarrow$

$\Rightarrow$ aria $(D) = \frac{\pi R^2}{2} = \frac{\pi}{8}$

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Facem schimbarea de variabilă și trecem la coord polare generalizate:

$$\begin{cases} x - \frac{1}{2} = r\cos t \\ y = r\sin t \end{cases} ; \; r \in [0, \tfrac{1}{2}] ; \; t \in [0, \pi] ; \; J(r,t) = r$$

$$\Rightarrow \iint_D f(x,y)\,dxdy = \int_0^{\pi} \left(\int_0^{1/2} \left(\frac{1}{2} + r\cos t\right) r\sin t \cdot r\,dr \right) dt =$$

$$= \int_0^{\pi} \left(\int_0^{1/2} \left(\frac{1}{2} r^2 \sin t + r^3 \sin t \cos t \right) dr \right) dt =$$

$$= \int_0^{\pi} \left(\frac{1}{6} r^3 \sin t + \frac{1}{4} r^4 \sin t \cos t \Big|_0^{1/2} \right) dt =$$

$$= \int_0^{\pi} \left(\frac{1}{24} \sin t + \frac{1}{64} \sin t \cos t \right) dt =$$

$$= \frac{1}{24} \int_0^{\pi} \sin t\,dt + \frac{1}{64} \int_0^{\pi} \sin t \cos t\,dt =$$

$$= \frac{1}{24} (-\cos t) \Big|_0^{\pi} + \frac{1}{64} \int_0^0 u\,du = \frac{1}{24}$$

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